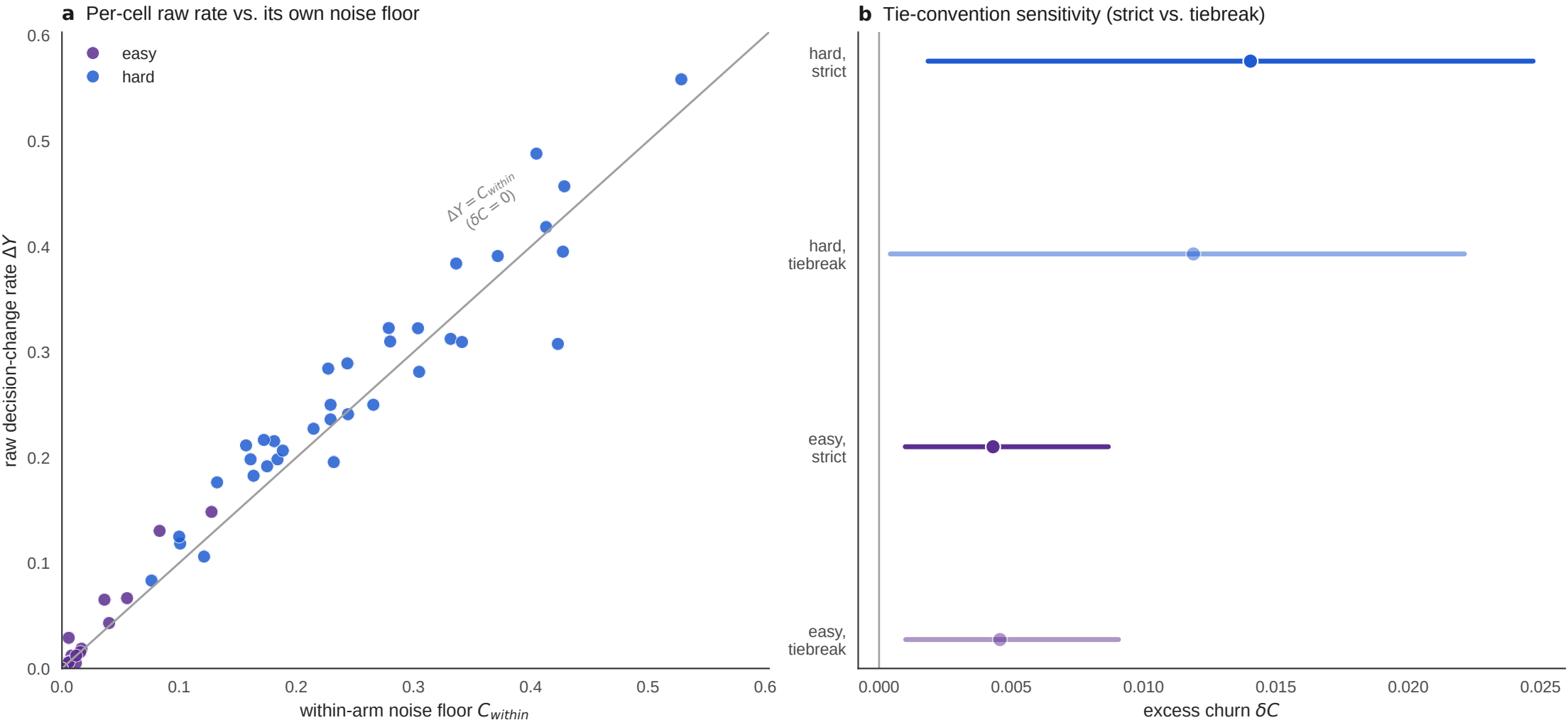Stochasticity atlas